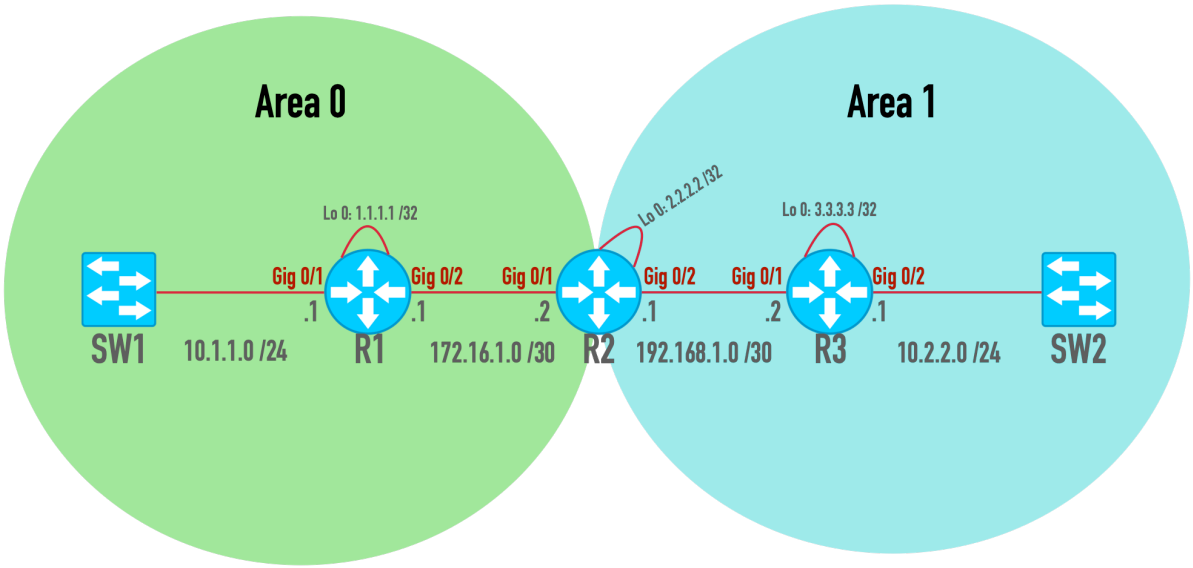
OSPFv2 Basic Configuration

Topology



Initial Configuration Commands

R1:

```
enable
conf t
host R1
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
exit
no ip-domain lookup
int gig 0/1
ip address 10.1.1.1 255.255.255.0
no shut
int gig 0/2
ip address 172.16.1.1 255.255.255.252
no shut
```

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```
int lo 0
ip address 1.1.1.1 255.255.255.255
end
copy run star
```

R2:

```
enable
conf t
host R2
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
exit
no ip-domain lookup
int gig 0/1
ip address 172.16.1.2 255.255.255.252
no shut
int gig 0/2
ip address 192.168.1.1 255.255.255.252
no shut
int lo 0
ip address 2.2.2.2 255.255.255.255
end
copy run star
```

R3:

```
enable
conf t
host R3
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
```

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```
line con 0
logging synchronous
exit
no ip-domain lookup
int gig 0/1
ip address 192.168.1.2 255.255.255.252
no shut
int gig 0/2
ip address 10.2.2.1 255.255.255.0
no shut
int lo 0
ip address 3.3.3.3 255.255.255.255
end
copy run star
```

Lab Tasks

1. Let's make all of our interfaces participate in OSPFv2.
2. Let's verify that OSPFv2 is set up correctly on R3 by using the "show ip route" command, pinging R1's loopback interface, and by using the "show ip protocols" command.
3. Let's verify that OSPFv2 is set up correctly on R2 by using the "show ip ospf neighbor" command, "show ip ospf rib" command, and the "show ip ospf interface" command on gig 0/1.
4. On R1, let's make gig 0/1 a passive interface.

Solution

Step 1: Let's make all of our interfaces participate in OSPFv2.

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#network 10.1.1.0 0.0.0.255 area 0
R1(config-router)#network 1.1.1.1 0.0.0.0 area 0
R1(config-router)#network 172.16.1.0 0.0.0.3 area 0
R1(config-router)#end
```

```
R2>en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 2
R2(config-router)#network 172.16.1.0 0.0.0.3 area 0
R2(config-router)#network 2.2.2.2 0.0.0.0 area 1
R2(config-router)#network 192.168.1.0 0.0.0.3 area 1
```

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```
R2(config-router)#end

R3>en
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#network 0.0.0.0 255.255.255.255 area 1
R3(config-router)#end
```

(##Notice on R3, that we gave a network statement that encompassed all IP addresses attached to this router.)

Step 2: Let's verify that OSPFv2 is set up correctly on R3 by using the "show ip route" command, pinging R1's loopback interface, and by using the "show ip protocols" command.

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
O IA     1.1.1.1 [110/3] via 192.168.1.1, 00:01:22, GigabitEthernet0/1
      2.0.0.0/32 is subnetted, 1 subnets
O        2.2.2.2 [110/2] via 192.168.1.1, 00:01:22, GigabitEthernet0/1
      3.0.0.0/32 is subnetted, 1 subnets
C        3.3.3.3 is directly connected, Loopback0
      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O IA     10.1.1.0/24 [110/3] via 192.168.1.1, 00:01:22, GigabitEthernet0/1
C        10.2.2.0/24 is directly connected, GigabitEthernet0/2
L        10.2.2.1/32 is directly connected, GigabitEthernet0/2
      172.16.0.0/30 is subnetted, 1 subnets
O IA     172.16.1.0 [110/2] via 192.168.1.1, 00:01:22, GigabitEthernet0/1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/30 is directly connected, GigabitEthernet0/1
L        192.168.1.2/32 is directly connected, GigabitEthernet0/1
R3#ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 7/10/15 ms
R3#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "application"
  Sending updates every 0 seconds
```

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```
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 3.3.3.3
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    0.0.0.0 255.255.255.255 area 1
  Routing Information Sources:
    Gateway         Distance      Last Update
    2.2.2.2              110      00:05:07
  Distance: (default is 110)
```

(##Notice how we recognize the loopback address of all of our routers interfaces.)
(##Notice how we are running OSPF process I.D. 1, our router I.D. is 3.3.3.3, and we are routing for all possible networks.)

Step 3: Let's verify that OSPFv2 is set up correctly on R2 by using the "show ip ospf neighbor" command, "show ip ospf rib" command, and the "show ip ospf interface" command on gig 0/1.

```
R2#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/DR         00:00:31    172.16.1.1      GigabitEthernet0/1
3.3.3.3           1   FULL/BDR        00:00:35    192.168.1.2     GigabitEthernet0/2
R2#show ip ospf rib

            OSPF Router with ID (2.2.2.2) (Process ID 2)


                Base Topology (MTID 0)

OSPF local RIB
Codes: * - Best, > - Installed in global RIB

*>  1.1.1.1/32, Intra, cost 2, area 0
      via 172.16.1.1, GigabitEthernet0/1
*   2.2.2.2/32, Intra, cost 1, area 1, Connected
      via 2.2.2.2, Loopback0
*>  3.3.3.3/32, Intra, cost 2, area 1
      via 192.168.1.2, GigabitEthernet0/2
*>  10.1.1.0/24, Intra, cost 2, area 0
      via 172.16.1.1, GigabitEthernet0/1
*>  10.2.2.0/24, Intra, cost 2, area 1
```

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```
      via 192.168.1.2, GigabitEthernet0/2
*   172.16.1.0/30, Intra, cost 1, area 0, Connected
      via 172.16.1.2, GigabitEthernet0/1
*   192.168.1.0/30, Intra, cost 1, area 1, Connected
      via 192.168.1.1, GigabitEthernet0/2
R2#show ip ospf int gig 0/1
GigabitEthernet0/1 is up, line protocol is up
  Internet Address 172.16.1.2/30, Area 0, Attached via Network Statement
  Process ID 2, Router ID 2.2.2.2, Network Type BROADCAST, Cost: 1
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           1         no          no            Base
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 172.16.1.1
  Backup Designated router (ID) 2.2.2.2, Interface address 172.16.1.2
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:05
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1/1, flood queue length 0
  Next 0x0(0)/0x0(0)/0x0(0)
  Last flood scan length is 2, maximum is 2
  Last flood scan time is 2 msec, maximum is 2 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 1.1.1.1  (Designated Router)
  Suppress hello for 0 neighbor(s)
```

(##Notice how we have a neighborship with R1 and a neighborship with R3.)

(##If we have more than one way, the best path will be indicated with an asterisks (*). The greater than sign (>) indicates that we have learned this route via OSPF.)

(##We see we have a cost of 1 going out of this interface because it is a gig interface.)

Step 4: On R1, let's make gig 0/1 a passive interface.

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#passive-interface gig 0/1
R1(config-router)#end
R1#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "application"
  Sending updates every 0 seconds
```

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```
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    1.1.1.1 0.0.0.0 area 0
    10.1.1.0 0.0.0.255 area 0
    172.16.1.0 0.0.0.3 area 0
  Passive Interface(s):
    GigabitEthernet0/1
  Routing Information Sources:
    Gateway         Distance      Last Update
    2.2.2.2              110      00:21:31
  Distance: (default is 110)
```

(##We are still advertising this route via OSPF, but we are not sending or receiving HELLO messages on that interface because it is directly connected to a switch.)

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